The Invariance of Incompressible Navier-Stokes Smoothness Under Conservative Linear Body Forces

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> **Axiom:** Let a fluid system exist within a domain $\Omega \subset \mathbb{R}^3$. The external body force acting on the system is restricted to a uniform, conservative gravitational field **f** = **g**.

1. The Governing Equations

We begin with the standard 3D Incompressible Navier-Stokes equations representing conservation of momentum and mass:

$$\partial u/\partial t + (u \cdot \nabla)u = -(1/\rho)\nabla p + \nu\nabla^2 u + g$$

$$\nabla \cdot u = 0$$

Where:

- **u**(x,y,z,t) is the velocity vector field.
- p(x,y,z,t) is the scalar pressure field.
- ρ is constant fluid density, and ν is the kinematic viscosity.
- **g** is the uniform acceleration vector due to gravity.

2. The Objective of the Proof

The objective is to prove that the presence or absence of a uniform gravitational field **g** has zero structural effect on the mathematical stability, turbulence development, or potential "blow-up" singularities of the velocity field **u**. This demonstrates that the core fluid dynamics are fundamentally uncoupled from external linear gravitational acceleration.

3. The Mathematical Transformation (The Zero-G Proof)

Because gravity **g** is uniform across space, it can be written as the gradient of a physical potential energy function, Φ:

$$\mathbf{g} = \nabla(\mathbf{g} \cdot \mathbf{x})$$

Where **x** = (x, y, z) is the spatial position vector. Substituting this potential representation back into the momentum equation yields:

$$\partial \mathbf{u}/\partial t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -(1/\rho)\nabla p + \nu\nabla^2\mathbf{u} + \nabla(\mathbf{g} \cdot \mathbf{x})$$

Utilizing the mathematical linearity of the gradient operator (∇), the internal fluid pressure gradient and the external gravitational potential gradient can be grouped together under a single spatial differential:

$$\partial \mathbf{u}/\partial t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla(p/\rho - \mathbf{g} \cdot \mathbf{x}) + \nu\nabla^2\mathbf{u}$$

We now introduce a redefined parameter designated as the **Modified Pressure** (P^*), which completely absorbs the gravitational potential energy field relative to the background framework:

$$P^* = p/\rho - \mathbf{g} \cdot \mathbf{x}$$

Substituting P^* back into the original momentum conservation equation yields the final transformed system:

$$\partial \mathbf{u}/\partial t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla P^* + \nu\nabla^2\mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

4. Conclusion & Q.E.D.

As demonstrated in the final transformed system, the explicit gravity vector **g** has been completely eliminated from the fluid's dynamic equations of motion. The velocity field vectors **u** (governing chaotic sloshing, turbulent eddies, and vortex behavior) are restricted entirely to the non-linear advection term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ and viscous diffusion $\nu\nabla^2\mathbf{u}$.

Altering the external acceleration field from 9.81 m/s^2 (Earth boundary) to 0 m/s^2 (Zero-G operational orbit) modifies only the background static reference pressure field (P^*). It exerts no dynamic influence on fluid particle acceleration or the structural existence of mathematical singularities.

Therefore, the existence of a smooth, well-behaved solution to the incompressible Navier-Stokes equations is fundamentally independent of gravity.

Q.E.D. (Quod Erat Demonstrandum)